

Organic chemistry

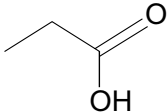
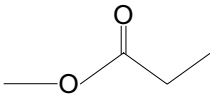
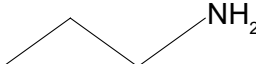
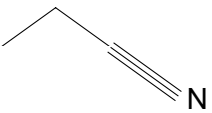
13 An introduction to AS Level organic chemistry

In this syllabus the following conventions are used:

X to represent a halogen atom

R and R' to represent alkyl groups (or in some circumstances a hydrogen atom); R and R' can be the same or different depending on the molecule.

homologous series	name of functional group	structural formula of functional group	displayed formula	skeletal formula	name
alkene	C=C bond	$\begin{array}{c} \text{R} \quad \text{R} \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{R} \quad \text{R} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \diagup \\ \text{H}-\text{C}-\text{C}=\text{C} \\ \quad \quad \\ \text{H} \quad \quad \text{H} \end{array}$		propene
halogenoalkane (primary, secondary and tertiary)	halogen	R—X	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{X} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$		1-chloropropane (when X is chlorine)
alcohol (primary, secondary and tertiary)	hydroxyl	R—OH	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$		propan-1-ol
aldehyde	carbonyl	$\begin{array}{c} \text{O} \\ \parallel \\ \text{R}-\text{C} \\ \diagup \\ \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \parallel \\ \text{H}-\text{C}-\text{C}-\text{C} \\ \quad \quad \diagup \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$		propanal
ketone	carbonyl	$\begin{array}{c} \text{R} \quad \text{O} \\ \diagdown \quad \parallel \\ \text{C} \\ \diagup \\ \text{R}' \end{array}$	$\begin{array}{c} \text{H} \quad \text{O} \quad \text{H} \\ \quad \parallel \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \quad \text{H} \end{array}$		propanone

homologous series	name of functional group	structural formula of functional group	displayed formula	skeletal formula	name
carboxylic acid	carboxyl	$\text{R}-\text{C} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{OH} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{C} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O}-\text{H} \end{array} \\ \quad \\ \text{H} \quad \text{H} \end{array}$		propanoic acid
ester	ester	$\text{R}-\text{C}-\text{O}-\text{C} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{R} \end{array}$	$\begin{array}{c} \text{H} \quad \text{O} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{O}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$		methyl propanoate
amine (primary only)	amine	$\text{R}-\text{NH}_2$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{N} \begin{array}{l} \nearrow \text{H} \\ \searrow \text{H} \end{array} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$		propylamine
nitrile	nitrile	$\text{R}-\text{C}\equiv\text{N}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{C}\equiv\text{N} \\ \quad \\ \text{H} \quad \text{H} \end{array}$		propanenitrile